

Optimal Local Edge-Set Compression in Regular Graphs

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Abstract

Fix a degree d . An omniscient encoder is given a finite simple d -regular graph, unique vertex identifiers, and one arbitrary bit on every edge. It writes the same number of bits at every vertex. A fixed decoder must recover the bit of a queried edge from a constant-radius neighborhood, where the radius may depend on d but not on the graph size. The decoder is not given the graph size, and identifiers are arbitrary distinct natural numbers. We determine the exact label length for every positive degree:

$$b_{\text{reg}}(d) = \lceil (d+1)/2 \rceil = \lfloor d/2 \rfloor + 1.$$

For odd d this is the integral counting bound. For even d , counting suggests $d/2$ bits, but uniform local decoding forces one additional bit.

The construction is explicit and self-contained. Star-shaped marker words expose bounded connected Voronoi blocks. A canonical in-block matching gives a self-synchronizing high-rate block code. After canceling parallel boundary edges in opposite pairs, paired incidences turn the remaining quotient into virtual paths and cycles. A small part of the block-code surplus stores occasional absolute direction records on these paths and cycles. A finite capacitated Hall argument places the records densely enough for local decoding, while the induced orientation makes every block own at most half of its boundary edges, up to rounding. No local orientation theorem, Lovász Local Lemma, or prior compression scheme is used as a black box. The even lower bound uses zero-slack circulant strips. Finite-ring surjectivity makes their induced maps reversible; reflection moves an odd total number of binary edge tracks, contradicting a self-contained rational information-flow index. With simple uniform constants the upper-bound radius is $O(d^5 \log d)$ for $d \geq 3$; for cubic graphs it is 1,007,136,282.

Contents

1	Introduction	2
2	Model and statement of the theorem	3
3	Proof idea	5
4	Self-exposing blocks and their code	6
5	From the block quotient to virtual strands	10
6	How many direction records fit?	11
7	Placing and locally reading the records	13

8	The encoder	16
9	A total local decoder	17
10	Correctness, locality, and the explicit radius	18
11	Order normalization and reversible strip maps	22
12	A self-contained rational flow index	25
13	The even-degree lower bound	27
14	Correctness and optimality	29

1 Introduction

Local computation usually asks what a network can compute after a bounded number of communication rounds. Here the network additionally receives a short, instance-dependent label at each vertex. The labels may be chosen by an unrestricted centralized encoder, but the decompressor must be one fixed local rule. This “local advice and decompression” viewpoint was formulated for graph problems by Balliu et al. [BBK⁺24, BBK⁺25]; it sits within the broader traditions of the LOCAL model [NS95] and distributed computation with advice [FGIP09].

We study the cleanest lossless compression problem in this model. Each edge of a known graph carries an arbitrary hidden bit—equivalently, the hidden object is an arbitrary subset of the edges. The topology and unique vertex identifiers are free side information. Labels are placed on vertices, and the bit of each edge must be readable nearby.

On an n -vertex d -regular graph there are $dn/2$ independent edge bits. Thus labels of s bits per vertex can work only if $sn \geq dn/2$. For odd d , integrality gives $s \geq (d+1)/2$. For even d , this elementary argument only gives $s \geq d/2$. Balliu et al. [BBK⁺25] gave a general upper bound one bit above the integral counting lower bound and singled out the cubic two-bit case. Our construction answers that cubic question. For the graph-size-independent decoding model defined below, we also determine the optimum in every regular degree: constructive entropy sharing settles the odd degrees, and a reversible one-dimensional index rules out the zero-slack rate in the positive even degrees.

Several neighboring source-coding models use similar terminology but differ from this worst-case graph-metric game. Makhdoumi et al. study recovery of source coordinates by probing selected coordinates of a globally compressed random sequence [MHMP15], while Vatedka, Chandar, and Tchamkerten study separately encoded correlated random sources [VCT22]. Delgosha and Anantharam study asymptotic compression of random marked sparse graphs and universal compression of the graph and its marks [DA22, DA20]. Here the graph is uncompressed side information, the edge bits are arbitrary, storage is uniformly bounded at every vertex, access is restricted to a bounded graph-distance view, and recovery is exact on every finite instance. Thus these works are adjacent context, not prior bounds for this model. Certificate-radius tradeoffs [FFH⁺21], distributed degree splitting [GHK⁺20], sinkless-orientation locality [BKK⁺23], and distributed versions of Hall’s theorem [BMN⁺25]

provide further methodological context; none is a proof dependency, and our matching use is ordinary finite Hall at the centralized encoder.

Main result, informally. For every $d \geq 1$, exactly $\lceil (d+1)/2 \rceil$ bits per vertex are necessary and sufficient to locally encode every edge subset of every finite simple d -regular graph. Odd degrees attain counting; even degrees require one bit more than counting.

The apparent obstacle is ownership. If every edge has one endpoint that “owns” it, that endpoint can store its bit. At the counting optimum, a block of b vertices has room for only about $db/2$ owned edges. Hence its boundary edges must be directed almost evenly in and out. A globally chosen balanced orientation is easy to imagine but is not free side information; the decoder must reconstruct it from the same optimal-length labels that carry the payload.

Our construction uses the at-least-half-bit-per-vertex margin between $d/2$ and $\lceil (d+1)/2 \rceil$. Large components are partitioned into bounded connected blocks, each marked by a recognizable star. The nonmarker labels form a product code with linear entropy surplus after reserving $\lceil db/2 \rceil$ payload bits. We use that surplus for sparse *direction records*. At the quotient level, a purely combinatorial incidence pairing decomposes the orientation problem into paths and cycles. One nearby record fixes the direction of an entire visible segment. A capacitated Hall argument guarantees that records can be stored by the blocks occurring in those segments. The Hall theorem needed for this step is proved in full in [Lemma 7.1](#).

The resulting proof is insensitive to expansion, girth, bridges, and block shape. A block may look like a tree, a ladder, or an arbitrary connected subgraph. Components too small to amortize a marker use a direct code, and a component contained in one large block simply has no quotient boundary.

Organization. [Section 2](#) fixes the exact decoding model and states the theorem. [Section 3](#) gives a visual proof overview. Self-exposing blocks and their finite code are built in [Section 4](#). The quotient orientation skeleton appears in [Section 5](#); record capacity and placement are proved in [Sections 6](#) and [7](#). The encoder and a total decoder on *all* labeled inputs are specified in [Sections 8](#) and [9](#). [Section 10](#) proves the radius bound. Order normalization, finite-ring reversibility, the rational flow index, and the even-degree obstruction are proved in [Sections 11](#) to [13](#). Finally, [Section 14](#) assembles the exact all-degree theorem.

2 Model and statement of the theorem

Input graphs are finite, undirected, and simple; they may be disconnected or empty. Infinite graphs occur only as auxiliary objects in the lower bound. Fix a degree d . An input consists of a d -regular graph $G = (V, E)$, an injective identifier map $\text{id} : V \rightarrow \mathbb{N}$, and a target vector $x \in \{0, 1\}^E$. Identifiers and the topology are visible to both encoder and decoder and are not charged to the label length. Whenever an order of incident edges is needed, we use increasing neighbor identifier, so no additional port numbering is required. Graph distances take the value $+\infty$ between different components.

For an edge $e = uv$, its radius- r edge ball has vertex set

$$V_r(G, e) = \{w \in V : \min\{\text{dist}(w, u), \text{dist}(w, v)\} \leq r\}.$$

For a labeling $\ell : V \rightarrow \Sigma$, write $\text{View}_r(G, \text{id}, \ell, e)$ for the finite rooted structure consisting of the edge e , the subgraph induced by $V_r(G, e)$, the restrictions of id and ℓ to this vertex set, and the following exact port table. The visible vertices are indexed canonically in increasing identifier order. At each visible vertex, the d ports are indexed by the identifier order of all its global neighbors. A port contains the corresponding local vertex when that neighbor is visible and an opaque stub otherwise. A stub reveals its port position, but no outside endpoint, identifier, label, or adjacency.

The decoder is a total function on the corresponding raw finite structures, not merely on structures that arise by extraction. Thus its input may have repeated identifiers or an inconsistent port table; the local graph is still simple, the root is still an actual local edge, and every label is still an element of Σ . Correctness is required only on the genuine extracted views just defined. This is the precise edge-view convention used throughout. Raw structures need not satisfy the extracted view's degree, size, connectedness, or root-distance constraints.

Definition 2.1 (Local edge-set code). Let Σ be a finite alphabet. A radius- r decoder is one fixed, deterministic, total function D that maps every Σ -labeled rooted radius- r edge view to a bit. It is valid for d -regular graphs if

$$\forall G \forall \text{id} \forall x \in \{0, 1\}^{E(G)} \exists \ell : V(G) \rightarrow \Sigma \forall e \in E(G) : D(\text{View}_r(G, \text{id}, \ell, e)) = x_e. \quad (2.1)$$

The encoder choosing ℓ is centralized and computationally unrestricted. It need not be local, canonical, or efficient. The decoder is not told the component size or the global number of vertices.

In particular, the quantifier order is $\exists r \exists D$ followed by the universal quantifiers in (2.1). The same D must work at every graph size and for every injective natural-number identifier assignment. This differs from versions of the LOCAL model in which vertices are supplied with the global size n or identifiers are restricted to a prescribed polynomial-size range. Our bit upper bound applies to those models as well: an endpoint can gather the edge view and the neighbor layer defining its ports in at most $r + 2$ communication rounds for a radius- r rule, then simulate the same edge-rooted decoder. The two endpoints obtain identical outputs. The counting lower bound still applies. The even-degree lower bound proved here is for (2.1); we do not claim its extension to a decoder supplied with n or to a restricted identifier range.

For $s \in \mathbb{N}$, labels of s bits mean an alphabet of exactly 2^s symbols. Let $b_{\text{reg}}(d)$ be the least s for which some finite radius and some decoder satisfy (2.1) for every finite simple d -regular graph. Equivalently, for each fixed graph and identifier assignment, the global map induced by applying D at every edge must be onto $\{0, 1\}^E$. This minimum exists: d bits always suffice by storing each incident edge bit in its port position and reading it at the lower-identifier endpoint. On raw views it breaks identifier ties by local vertex index, uses the first port pointing to the other root endpoint, and returns zero if no such port exists.

All logarithms below are base two. For every $d \geq 3$, define

$$\begin{aligned} s_d &= \left\lceil \frac{d+1}{2} \right\rceil, & \alpha_d &= 2^{s_d}, \\ \Sigma_d &= \{0, 1, \dots, \alpha_d - 1\}. \end{aligned} \quad (2.2)$$

$$b_\star = 64(d+1)^2, \quad \eta = 2b_\star - 2, \quad (2.3)$$

$$B_{\max} = 1 + d \sum_{j=0}^{\eta-1} (d-1)^j = 1 + \frac{d}{d-2} ((d-1)^\eta - 1), \quad (2.4)$$

$$\chi_d = \lceil \log_2(dB_{\max}) \rceil, \quad L = 20d\chi_d, \quad (2.5)$$

$$R_d = (2L + 3)(2\eta + 2). \quad (2.6)$$

The letter L is the length of a virtual record window, not the length of a physical label. The constants depend only on d ; their roles are summarized below.

parameter	purpose
$b_\star$	minimum size at which a marker is amortized
η	domination radius and malformed-input cutoff
$B_{\max}$	uniform upper bound on a parsed block size
χ_d	uniform upper bound on a record width
L	length of a virtual window
R_d	advertised physical decoding radius

Theorem 2.2 (Optimal compression in every regular degree). *For every $d \geq 3$, there is a deterministic radius- R_d decoder over Σ_d satisfying (2.1). Consequently,*

$$b_{\text{reg}}(d) = \left\lceil \frac{d+1}{2} \right\rceil = \left\lfloor \frac{d}{2} \right\rfloor + 1 \quad (d \geq 1). \quad (2.7)$$

For $d = 3$, the displayed construction has

$$R_3 = 1,007,136,282. \quad (2.8)$$

The edgeless case has $b_{\text{reg}}(0) = 0$.

The upper bounds for $d = 1, 2$ use the direct codes described in [Section 14](#); the even-degree lower-bound argument also includes $d = 2$. In the upper-bound construction below, $d \geq 3$ and all constants are those in (2.2)–(2.6). Ceilings of logarithms may always be evaluated by integer comparisons; no real arithmetic is required of the decoder.

3 Proof idea

The information budget is easiest to see blockwise. Let $E(B)$ be the set of edges with both endpoints in a block B , let ∂B be its edge boundary, and put $b = |B|$. Regularity gives $db = 2|E(B)| + |\partial B|$. The average target-bit budget at the counting rate is $db/2$. If B owns every internal edge and at most half of its boundary edges (rounded up), then it owns at most

$$|E(B)| + \left\lceil \frac{|\partial B|}{2} \right\rceil = \left\lceil \frac{db}{2} \right\rceil$$

edges. These are the payload bits that B must store.

[Figure 1](#) summarizes the construction. Every step is locally reconstructible on a valid encoding. In particular, the decoder discovers the blocks from the physical labels before it interprets their logical contents, so there is no circular use of hidden metadata.

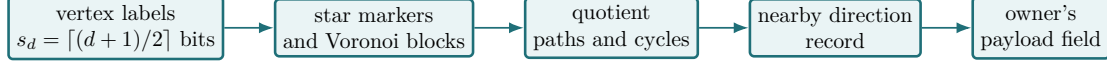


Figure 1: The decoder's dependency chain. Window boundaries and the Hall assignment are encoder-side devices; the decoder only needs their explicit nearby records.

Why is there room for structural information? The marker uses a bounded number of vertices in a large block. On the other vertices, even the restricted alphabet $\Sigma_d \setminus \{1\}$ carries $\log_2(2^{s_d} - 1) > d/2$ bits per vertex. After marker losses and integer rounding, we retain at least $b/20$ extra logical bits in every block. These bits are split into record slots.

To balance boundary ownership, parallel quotient edges are first canceled in oppositely directed pairs. The remaining quotient is simple. At every quotient vertex, pair consecutive incidences. Splitting a vertex according to these pairs turns the quotient into disjoint paths and cycles. Directing each path or cycle consistently makes every paired copy have one incoming and one outgoing edge; only a possible singleton can be unbalanced. Sparse absolute records make this direction locally recoverable. The capacity argument is global only at encoding time: Hall's theorem assigns each length- L path window to a record slot of a block occurring in it.

4 Self-exposing blocks and their code

We begin with a connected component containing at least $b_\star$ vertices. Small components will be encoded directly in [Section 8](#).

4.1 Bounded connected Voronoi blocks

The encoder chooses an inclusion-maximal set $\mathcal{C}$ of vertices whose distinct members have pairwise distance at least

$$2b_\star - 1 = \eta + 1. \quad (4.1)$$

Such a set exists because the component is finite. Maximality implies that every vertex is within distance η of $\mathcal{C}$: otherwise that vertex could be added. Assign each vertex v to the center $c \in \mathcal{C}$ that minimizes

$$(\text{dist}(v, c), \text{id}(c)) \quad (4.2)$$

lexicographically. Write B_c for the resulting block.

Lemma 4.1 (Block geometry). *Every B_c is connected, contains the closed neighborhood $N[c]$, and has*

$$b_\star \leq |B_c| \leq B_{\max}. \quad (4.3)$$

Proof. Domination puts B_c inside the radius- η ball about c . A graph of maximum degree d has at most $1 + d \sum_{j=0}^{\eta-1} (d-1)^j = B_{\max}$ vertices in such a ball, proving the upper bound: a shortest path has at most d choices for its first edge and at most $d-1$ choices at each later step, since it cannot immediately return along the preceding edge.

If $\text{dist}(v, c) \leq b_\star - 1$ and $c' \neq c$ is another center, then

$$\text{dist}(v, c') \geq \text{dist}(c, c') - \text{dist}(v, c) \geq (2b_\star - 1) - (b_\star - 1) = b_\star.$$

Thus v is strictly closer to c than to every other center. The entire radius- $(b_\star - 1)$ ball about c lies in B_c . This ball has at least $b_\star$ vertices: if a geodesic of length $b_\star - 1$ exists, take its vertices; otherwise the ball is the whole component, which by assumption has at least $b_\star$ vertices. This also gives $N[c] \subseteq B_c$.

For connectedness, let $v \in B_c$, and let w be the next vertex on a shortest v - c path. If a center c' beat c in the lexicographic rule at w , then $\text{dist}(v, c') \leq \text{dist}(w, c') + 1$. A strict distance advantage at w remains an advantage at v ; in the tied-distance case, the same identifier tie-break also makes c' beat c at v . Both contradict $v \in B_c$. Hence $w \in B_c$, and iteration gives a path from v to c inside B_c . $\square$

The argument includes the case in which the entire component is one block. No growth or expansion assumption was used.

4.2 A marker that cannot be forged by data

Set $\alpha = \alpha_d$ and $\Sigma = \Sigma_d$ for brevity. A vertex is called a *marked center* when its label is 1 and all its d neighbors have label 0. In an intended block $B = B_c$, the encoder fixes label 1 at c and label 0 at every vertex in $N(c)$. The data region is

$$F_B = B \setminus N[c], \quad r_B = |F_B| = |B| - d - 1. \quad (4.4)$$

Order all graph edges by the globally unique key

$$\kappa(uv) = (\min\{\text{id}(u), \text{id}(v)\}, \max\{\text{id}(u), \text{id}(v)\}). \quad (4.5)$$

Run the greedy matching algorithm on the induced graph $G[F_B]$, processing edges in increasing κ order. This gives a canonical maximal *buddy matching* M_B ; put $m_B = |M_B|$. Order the endpoints of every matched edge by identifier. A word on F_B is *legal* when

- an unmatched vertex receives any symbol except 1; and
- a matched ordered pair receives any element of Σ^2 except $(1, 0)$ and $(0, 1)$.

Therefore the exact number of legal data words is

$$N_B = (\alpha^2 - 2)^{m_B} (\alpha - 1)^{r_B - 2m_B}. \quad (4.6)$$

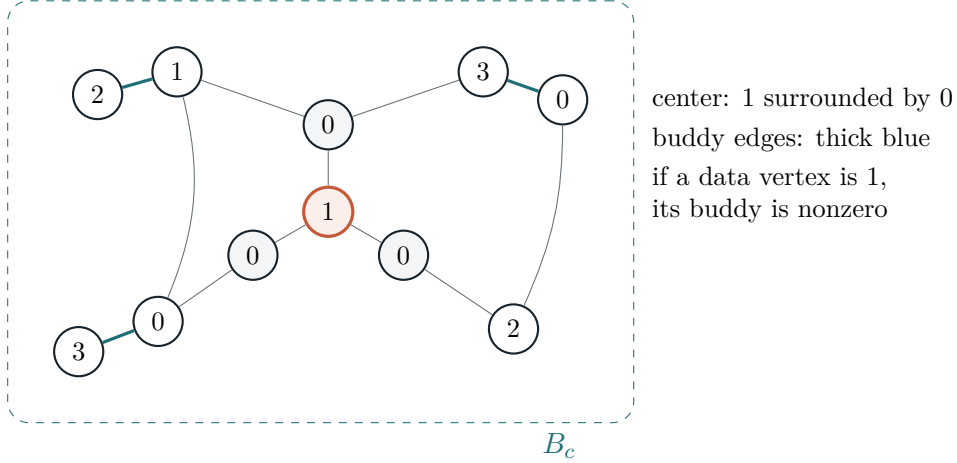


Figure 2: A schematic cubic block, where $\Sigma = \{0, 1, 2, 3\}$. Only edges relevant to the marker and sample buddy pairs are shown; omitted incidences continue inside or outside the block. The drawing is not to scale: an actual large block has at least 1024 vertices. The displayed data labels are one legal word. On each buddy edge all 16 ordered pairs except $(1, 0)$ and $(0, 1)$ are permitted.

Lemma 4.2 (Self-synchronization and code size). *In a legal block word, the center c is the only marked center in B . Moreover,*

$$N_B \geq (\alpha - 1)^{|B|-d-1}. \quad (4.7)$$

Proof. The vertex c is marked by construction, and its neighbors have label zero. An unmatched data vertex cannot have label 1. If a matched data vertex has label 1, its buddy is an actual graph neighbor and is nonzero, by the two forbidden pairs. It therefore cannot be marked. These cases cover B .

For the count, observe that

$$\alpha^2 - 2 - (\alpha - 1)^2 = 2\alpha - 3 > 0$$

because $\alpha \geq 4$. Replacing every matched-pair factor in (4.6) by $(\alpha - 1)^2$ proves (4.7). $\square$

Fix once and for all increasing orders of the $\alpha - 1$ unmatched symbols and lexicographic order of the $\alpha^2 - 2$ legal ordered pairs. Together with increasing identifier order of unmatched vertices and increasing κ order of matching edges, the block topology and identifiers determine a mixed-radix enumeration

$$\{0, 1, \dots, N_B - 1\} \longleftrightarrow \{\text{legal words on } F_B\}. \quad (4.8)$$

For definiteness, list the matched-edge coordinates first and the unmatched vertices second. If their radices are $r_0, \dots, r_{u-1}$ and their zero-based symbol ranks are $a_0, \dots, a_{u-1}$, the word rank is $\sum_{i=0}^{u-1} a_i \prod_{j<i} r_j$. Successive Euclidean divisions give the inverse map. Empty products equal one and empty sums equal zero. Unranking an integer through this enumeration is a finite, explicit operation on the topology and identifiers. Ranking is its inverse and additionally reads the physical symbols on F_B .

Applying Lemma 4.2 to all blocks shows that the marked centers in the completed physical labeling are *exactly* $\mathcal{C}$. Hence a decoder first detects the stars, then repeats (4.2), and reconstructs the same blocks and buddy matchings. The logical contents of (4.8) are not used to discover the blocks.

4.3 Linear structural surplus

For a block of size $b = |B|$, reserve

$$\pi_B = \left\lceil \frac{db}{2} \right\rceil \quad (4.9)$$

payload coordinates. We use the following conservative whole-bit budget and its remaining structural surplus:

$$c_B = \left\lfloor \log_2((\alpha - 1)^{b-d-1}) \right\rfloor, \quad \sigma_B = c_B - \pi_B. \quad (4.10)$$

By (4.7), $2^{c_B} \leq N_B$. The exact legal-word count N_B is used for physical ranking and unranking, but all field lengths use the budget c_B in (4.10), not $\lfloor \log_2 N_B \rfloor$. Thus the field lengths depend only on d , b , and the residual degree defined below.

Lemma 4.3 (Uniform surplus). *Every block produced above satisfies*

$$\sigma_B \geq \frac{|B|}{20}. \quad (4.11)$$

Proof. Write $s = s_d = \lceil (d+1)/2 \rceil$. On integers $s \geq 2$, the function

$$f(s) = \log_2(2^s - 1) - (s - 1/2)$$

is increasing, because

$$f(s+1) - f(s) = \log_2\left(\frac{2^{s+1} - 1}{2(2^s - 1)}\right) > 0.$$

Also $3^{16} = 43,046,721 > 33,554,432 = 2^{25}$, so $f(2) > 1/16$. Since $s \geq (d+1)/2$, this yields, for every $d \geq 3$,

$$\log_2(\alpha - 1) \geq \frac{d}{2} + \frac{1}{16}. \quad (4.12)$$

By (4.10), $\lfloor y \rfloor \geq y - 1$, and $\lceil db/2 \rceil \leq db/2 + 1/2$,

$$\begin{aligned} \sigma_B &\geq (b - d - 1) \left(\frac{d}{2} + \frac{1}{16} \right) - 1 - \frac{db}{2} - \frac{1}{2} \\ &= \frac{b}{16} - (d+1) \left(\frac{d}{2} + \frac{1}{16} \right) - \frac{3}{2} \\ &\geq \frac{b}{16} - \frac{(d+1)^2}{2} - \frac{3}{2}. \end{aligned}$$

Because $b \geq b_\star = 64(d+1)^2$, the middle loss is at most $b/128$. Furthermore $b \geq 1024 > 320$, so $3/2 \leq 3b/640$. Consequently

$$\sigma_B \geq \frac{b}{16} - \frac{b}{128} - \frac{3b}{640} = \frac{b}{20}.$$

□

The budget deliberately leaves the positive rate improvement from buddy pairs unused. It therefore covers every connected block shape, even one whose data region has a very small matching. The matching permits symbol 1 at matched data vertices without creating false markers; the resulting larger physical code is still enumerated exactly by (4.8).

5 From the block quotient to virtual strands

Fix the encoder-produced blocks in one large connected component. The quotient multigraph J has one vertex for every block. Every original edge whose endpoints lie in different blocks gives one edge of J ; internal edges do not enter J and hence create no loops.

For the rest of the construction, write z_B for the marked center of a physical block B .

For each unordered pair of blocks, sort its parallel quotient edges by the original-edge key κ from (4.5), pair consecutive edges, and delete each pair from the residual quotient. In a deleted pair, direct the first edge from the lower-identifier center to the higher-identifier center and the second in the reverse direction. If the multiplicity is odd, its last edge remains. Thus the residual quotient K is finite, simple, and loopless. Isolated quotient vertices are allowed.

Let k_B be the degree of B in K . Since K is simple, the identifiers of the centers at the other endpoints give a tie-free order of the incidences at B . Pair incidences $(1, 2), (3, 4), \dots$, leaving the last one single if k_B is odd. Split B into one *virtual copy* for every pair or singleton. Each edge of K now joins the copies containing its two incidences. Call the resulting graph H . The number of copies above B is

$$\nu_B = \left\lceil \frac{k_B}{2} \right\rceil. \quad (5.1)$$

Index them in incidence order, and give the j th copy the key $(\text{id}(z_B), j)$. Paired copies have degree two and singleton copies have degree one; if $k_B = 0$, there is no copy. Hence H is a disjoint union of finite paths and cycles. Indeed, a walk starting at a degree-one vertex has a unique continuation at each internal vertex and ends at the other degree-one vertex; connectedness leaves no additional vertices in that component. A component without a degree-one vertex is finite and two-regular, so following successive edges returns to the starting vertex and exhausts a cycle. The same physical block may occur through several virtual copies, possibly on the same component of H . We call the components of H *virtual strands*.

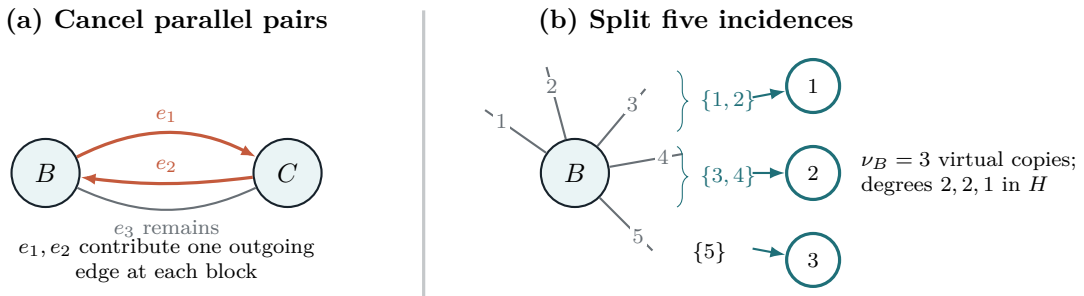


Figure 3: The quotient reduction. Oppositely directed parallel pairs are already balanced. Pairing the remaining incidences turns the residual orientation problem into paths and cycles.

A *consistent direction* of an H -path directs all edges from one endpoint to the other; on an H -cycle it chooses one of the two cyclic directions. Transferring these directions back to K gives the following elementary bound.

Lemma 5.1 (Balanced quotient ownership). *If every component of H is directed consistently,*

then

$$\text{outdeg}_K(B) \leq \nu_B = \left\lceil \frac{k_B}{2} \right\rceil. \quad (5.2)$$

After the canceled pairs are restored, direct each original interblock edge as above or as prescribed by K . If B owns its internal edges and its outgoing interblock edges, then it owns at most $\pi_B = \lceil d|B|/2 \rceil$ edges. Every original edge has exactly one owner.

Proof. At a paired virtual copy, a consistent path or cycle direction has one incoming and one outgoing incidence. Only a singleton copy can contribute an unmatched outgoing incidence. Thus every copy contributes at most one outgoing edge at its physical block, proving (5.2).

Let i_B and δ_B be the numbers of internal and boundary edges of B . Removing parallel pairs deletes an even number of incidences at B , so $\delta_B \equiv k_B \pmod{2}$. The removed pairs contribute exactly $(\delta_B - k_B)/2$ outgoing edges at B . Therefore the total number of outgoing boundary edges is at most

$$\frac{\delta_B - k_B}{2} + \left\lceil \frac{k_B}{2} \right\rceil = \left\lceil \frac{\delta_B}{2} \right\rceil. \quad (5.3)$$

Regularity gives $2i_B + \delta_B = d|B|$, and hence

$$i_B + \left\lceil \frac{\delta_B}{2} \right\rceil = \left\lceil \frac{d|B|}{2} \right\rceil = \pi_B. \quad (5.4)$$

Internal edges have their unique block as owner, and every interblock edge has exactly one tail, so ownership partitions $E(G)$. $\square$

Bridges merely appear on virtual paths and need no special treatment. If a whole component is one block, then K and H are empty and the lemma says that all internal edges fit in the ordinary payload.

6 How many direction records fit?

The orientation of H has not yet been made locally visible. A record stored by B will be either empty or will name one of its ν_B virtual copies and specify whether a fixed reference incidence at that copy points out of or into B . For a paired copy, take its first incidence in the local order as reference; for a singleton, take its unique incidence. Thus there are $2\nu_B + 1$ record states.

When $k_B > 0$, define the record width and number of available slots by

$$w_B = \lceil \log_2(2\nu_B + 1) \rceil, \quad t_B = \left\lfloor \frac{\sigma_B}{w_B} \right\rfloor. \quad (6.1)$$

When $k_B = 0$, set $\nu_B = w_B = t_B = 0$. In w_B bits, value 0 means empty; for $1 \leq j \leq \nu_B$, the values $2j - 1$ and $2j$ mean, respectively, that copy j 's reference incidence is incoming and outgoing. Any remaining bit patterns are invalid and will also be read as empty on malformed inputs.

Example 6.1 (A residual degree-five block). If $k_B = 5$, then $\nu_B = 3$ and $w_B = \lceil \log_2 7 \rceil = 3$. A slot has exactly the seven semantic states

$$\text{empty}, \quad (1, \text{in}), (1, \text{out}), \quad (2, \text{in}), (2, \text{out}), \quad (3, \text{in}), (3, \text{out}),$$

plus one unused three-bit pattern. The record contains an absolute local direction, not a fragile bit whose meaning depends on an unseen global orientation.

Lemma 6.2 (Record-slot capacity). *Let B be any connected block with $b_\star \leq |B| \leq B_{\max}$ that contains the closed neighborhood of its designated center and is equipped with the code above. Then*

$$\nu_B \leq Lt_B. \quad (6.2)$$

Moreover, all t_B record fields and all π_B payload fields fit in its code:

$$\pi_B + t_B w_B \leq c_B. \quad (6.3)$$

Proof. The case $k_B = 0$ is immediate, so suppose $k_B > 0$ and write $b = |B|$. Connectedness gives $i_B \geq b - 1$, whence

$$\delta_B = db - 2i_B \leq (d - 2)b + 2, \quad k_B \leq \delta_B. \quad (6.4)$$

From $\nu_B = \lceil k_B/2 \rceil$,

$$2\nu_B + 1 \leq k_B + 2 \leq (d - 2)b + 4 \leq db, \quad (6.5)$$

where the last inequality uses $b \geq 2$. Consequently $w_B \leq \lceil \log_2(db) \rceil$.

We next prove the deliberately loose but useful estimate

$$w_B \leq \frac{b}{40}. \quad (6.6)$$

Since $b \geq 64(d + 1)^2$, we have $d < \sqrt{b}/8$, and hence

$$\log_2(db) < \frac{3}{2} \log_2 b - 3.$$

Taking a ceiling gives

$$w_B < \frac{3}{2} \log_2 b - 2.$$

For $b \geq 1024$, the function $b/40 - (3/2) \log_2 b + 2$ is positive at 1024 and has positive derivative; thus (6.6) follows.

By Lemma 4.3, $\sigma_B/w_B \geq 2$. The elementary inequality $\lfloor y \rfloor \geq y/2$ for $y \geq 2$ gives

$$t_B \geq \frac{\sigma_B}{2w_B} > 0. \quad (6.7)$$

Also (6.5) implies $\nu_B < db/2$. Therefore

$$\frac{\nu_B}{t_B} \leq \frac{2\nu_B w_B}{\sigma_B} < 20dw_B. \quad (6.8)$$

Every block has $b \leq B_{\max}$, so $w_B \leq \lceil \log_2(db) \rceil \leq \chi_d$. Equations (2.5) and (6.8) yield $\nu_B/t_B < L$, which implies the integer inequality (6.2).

Finally, the definition of t_B gives $t_B w_B \leq \sigma_B$. Together with $c_B = \pi_B + \sigma_B$, this is (6.3). $\square$

Let

$$\ell_B = \pi_B + t_B w_B. \quad (6.9)$$

By (6.3), $2^{\ell_B} \leq 2^{c_B} \leq N_B$. If the payload bits are $p_0, \dots, p_{\pi_B-1}$ and the numerical record values are $v_0, \dots, v_{t_B-1}$, define the logical integer

$$W_B = \sum_{i=0}^{\pi_B-1} p_i 2^i + \sum_{j=0}^{t_B-1} v_j 2^{\pi_B+jw_B}.$$

Thus payload occupies the low bits and record slots follow in increasing order of significance. Every choice of ℓ_B logical bits gives $0 \leq W_B < 2^{\ell_B}$. Unrank W_B through (4.8) to obtain the physical word. Ranking followed by binary digit extraction reverses the map. A legal physical word of rank at least 2^{ℓ_B} , or an illegal physical word, is assigned the all-zero logical string. No separate validity flag is needed: this is simply the total malformed-word fallback, while every encoder-produced word lies in the reserved interval and is recovered exactly.

7 Placing and locally reading the records

Call a component of H *short* if it has fewer than L virtual vertices. It needs no record. Define its canonical coherent orientation without choosing a path or cycle presentation: take the least residual edge of the component in the order of its original-edge key κ , direct it from its lower-key virtual endpoint to its higher-key endpoint, and extend uniquely and consistently through the path or cycle. A decoder that sees the whole component recovers exactly this convention. The same canonical coherent orientation will be the total fallback when a malformed nonshort component has no usable record.

Now consider a component with $N \geq L$ virtual vertices. Give it a linear indexing. A path starts at its lower-key endpoint. A cycle starts at its least-key vertex, with the lower-key of its two neighbors as the next vertex. Partition the first $\lfloor N/L \rfloor L$ positions into disjoint consecutive *windows* of exactly L virtual vertices. The final remainder has fewer than L vertices. This indexing and its window boundaries are used only by the encoder.

Each window must be assigned to a physical block that occurs among its virtual vertices, and B may receive at most t_B windows. We prove the finite matching fact that makes this possible.

Lemma 7.1 (Finite capacitated Hall theorem). *Let $\mathcal{W}$ and $\mathcal{B}$ be finite sets. Each $W \in \mathcal{W}$ has a set of allowed neighbors in $\mathcal{B}$, and each $B \in \mathcal{B}$ has an integer capacity $t_B \geq 0$. There is an assignment of every W to an allowed neighbor, using B at most t_B times, provided that for every $\mathcal{X} \subseteq \mathcal{W}$,*

$$|\mathcal{X}| \leq \sum_{B \in N(\mathcal{X})} t_B. \quad (7.1)$$

Proof. Replace B by t_B separate clones, each adjacent to the same elements of $\mathcal{W}$. It suffices to prove that the resulting bipartite graph has a matching saturating $\mathcal{W}$.

Take a maximum matching. If some left vertex is unmatched, start from all unmatched left vertices and follow alternating paths: nonmatching edges from left to right, then matching edges from right to left. Let X and Y be the reached left and right vertices. Every neighbor of X lies in Y : an unmatched edge is traversed forward, while the matched neighbor of a reached nonroot left vertex is the vertex from which it was reached. No vertex of Y is unmatched: an alternating path to such a vertex would have unmatched endpoints, and flipping the matching status along the path would increase the matching size. Thus each vertex of Y is matched to a distinct vertex of X , while X also contains an unmatched

root. Thus $|Y| < |X|$, contradicting Hall's condition. The maximum matching therefore saturates the left side. Collapsing clones gives the required capacitated assignment. This is the standard finite Hall argument [Hal35], included here so that no matching theorem is used as a black box. $\square$

Lemma 7.2 (Window allocation). *All long-component windows can be assigned to blocks occurring in them, with no block B assigned more than t_B windows.*

Proof. For a family $\mathcal{X}$ of windows, let $n_B(\mathcal{X})$ be the number of virtual occurrences over B in those windows. The windows are globally vertex-disjoint and each has L vertices, so

$$L|\mathcal{X}| = \sum_B n_B(\mathcal{X}). \quad (7.2)$$

There are only ν_B virtual copies above B in all of H . By Lemma 6.2,

$$n_B(\mathcal{X}) \leq \nu_B \leq Lt_B. \quad (7.3)$$

Only a block occurring in a window of $\mathcal{X}$ has a nonzero summand. Summing (7.3) over those blocks and dividing (7.2) by L gives (7.1). Lemma 7.1 now applies. $\square$

Fix one such assignment. If a window is assigned to B , choose the least-key virtual occurrence above B in that window and store the record in the first unused slot of B . If several windows are assigned to the same physical block, their selected copy indices are distinct because the windows are disjoint; the records themselves occupy distinct slots by the capacity constraint.

Direct each long path or cycle in the positive direction of its linear indexing. The record at a selected copy names that local copy and states whether its fixed reference incidence is outgoing or incoming. Given one such absolute statement, there is a unique way to propagate a consistent direction along the containing path or cycle. All records on that component therefore agree.

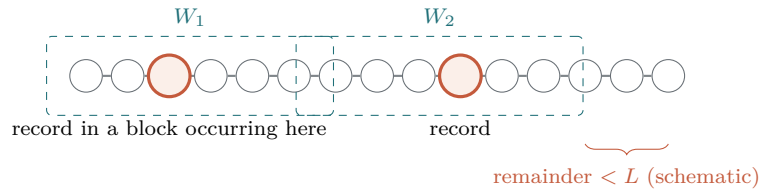


Figure 4: Record placement on a long virtual path (drawn with small schematic windows). Hall's theorem chooses the host block inside each window. The decoder does not reconstruct the windows; it searches for a named record.

For a virtual edge f and virtual vertex z , define $\text{dist}_H(f, z)$ as the minimum distance from z to an endpoint of f .

Lemma 7.3 (Record coverage). *Every edge of a long component of H is at distance less than $2L$ from a selected virtual copy carrying a record.*

Proof. Within a full window, every edge is fewer than L steps from that window's selected copy. The same is true at a boundary between consecutive full windows using a selected copy from one of the two adjacent windows.

For a path, only a terminal remainder exists. From the selected copy in the last full window to the end of the remainder is fewer than $2L$ steps. For a cycle, the remainder lies on the wraparound arc between the last and first full windows. Every edge on that arc is fewer than $2L$ steps from the selected copy in at least one of those two windows. When $N = L$, the one window covers all vertices; the closing cycle edge is also fewer than L steps from its selected copy. These cases include singleton path endpoints and an empty remainder. $\square$

7.1 The residual-orientation rule

We can now define the decoder's orientation of a queried residual edge f in the parsed virtual graph $\hat{H}$. The parser constructing $\hat{H}$ from a standalone finite view is specified in [Section 9](#); the present rule is applied only to that graph, not to an unavailable ambient component. Its paths, distances, and connected components in the following three steps are all computed in $\hat{H}$. The view is rooted at the original query edge e . Put

$$\rho_{\text{host}} = \eta + (2L + 1)(2\eta + 1). \quad (7.4)$$

A parsed record is *host-safe* if its named copy's physical host center has distance at most ρ_{host} from e in the supplied physical view graph. Of the two block incidences of a residual edge f , take the one at the lower center key (using the same malformed-input tie-break as elsewhere); let $a(f)$ be the virtual copy containing that incidence. This is the canonical anchor of f .

1. Compute the connected component of f in the finite standalone parsed virtual graph. If its vertex count is below L , use the canonical short-component direction.
2. Otherwise enumerate the records of all valid parsed cells. First treat every non-host-safe record as empty. Only after this safety filter is a remaining record eligible: its *named copy* (B, j) must lie in the same $\hat{H}$ -component and satisfy $\text{dist}_{\hat{H}}((B, j), a(f)) \leq 2L + 1$. Thus a slot stored at B but naming a different copy of B on another strand or outside this interval is ignored. Choose the eligible semantic record with least lexicographic key

$$((\text{id}(z_B), \text{visible-vertex tie-break}), j, \text{direction bit}),$$

read the absolute direction of its reference incidence, and propagate the unique consistent direction to f . Slot provenance has already been discarded; equal semantic records from several slots are one candidate.

3. If there is no eligible record, use the same canonical coherent component orientation as in the short case: the component's least residual edge points from its lower-key to its higher-key virtual endpoint, and this direction is propagated to f .

Invalid record values are empty. Duplicate or disagreeing records are resolved by the least-key rule. In particular, a syntactically valid record at an unsafe outer-boundary host is empty *before* component membership, distance, or minimum-key selection is considered. Thus this prescription gives one absolute direction to every residual edge on every parsed labeling, even a malformed one.

Correctness for the interior incidences needed by an original query is proved in [Lemma 10.4](#), after establishing exact reconstruction from the available physical ball. The encoder itself uses its chosen global strand directions and does not run this parser.

The distinction between a physical block and a named virtual copy is essential: one block can occur several times in H . The explicit filter in step 2 is what makes repeated occurrences harmless.

Example 7.4 (Why a record names a copy). Suppose one physical block B has copies $(B, 1)$ and $(B, 2)$ on two different virtual strands. A slot physically stored in B may name $(B, 2)$ and record the direction of its reference incidence. A query on the strand through $(B, 1)$ may still encounter the physical block B , but that record carries no orientation information about $(B, 1)$ and is ignored. A query near $(B, 2)$ accepts it. Merely naming the host block would conflate the two strands and could destroy the balance guarantee.

8 The encoder

We now encode an arbitrary target vector $x \in \{0, 1\}^{E(G)}$. Operations are performed independently in each connected component.

8.1 Large components

For a component with at least $b_\star$ vertices, the encoder performs the following finite operations.

1. Choose a maximal separated center set, form its Voronoi blocks, and construct each canonical buddy matching.
2. Construct J , cancel parallel pairs to obtain K , split incidences to obtain H , form the long-component windows, and choose a Hall assignment from [Lemma 7.2](#).
3. Orient the short strands canonically and the long strands in their positive indexed direction. Write their records into the assigned slots; fill unused slots with the empty value.
4. Restore the canceled quotient pairs and give every edge the owner from [Lemma 5.1](#). Sort the edges owned by B by κ , write their target bits into its π_B payload fields, and pad unused fields with zero.
5. Assemble the logical integer W_B and apply the explicit unranking map following (6.9). Put the resulting symbols on F_B , with the fixed center star on $N[z_B]$.

The ownership lemma shows that every target coordinate fits, while [Lemma 6.2](#) shows that every logical block word has a physical realization. Every edge is written exactly once.

All choices can, if desired, be made canonical by taking the lexicographically first maximal center set and Hall assignment. Their computational cost is irrelevant: the model restricts the decoder, not the encoder.

8.2 Small components

Let a component have $n < b_\star$ vertices. Every vertex can see the entire component within radius $b_\star - 1$. Order its vertices by identifier and its edges by κ , writing them as $v_0, \dots, v_{n-1}$ and $e_0, \dots, e_{m-1}$. Encode the integer $T = \sum_{j=0}^{m-1} x_{e_j} 2^j$ by its n base- α digits: $\ell(v_i) = \lfloor T/\alpha^i \rfloor \bmod \alpha$. This fits because

$$\frac{dn}{2} \leq s_d n. \tag{8.1}$$

The inverse forms $Z = \sum_i \ell(v_i) \alpha^i$ and, if $Z < 2^m$, reads the bit $\lfloor Z/2^j \rfloor \bmod 2$ at edge e_j ; otherwise it returns zero. On an arbitrary standalone view with $k < b_\star$ visible vertices, use these same formulas on the whole visible induced graph. If IDs repeat, use the total orders specified in [Section 9](#). No inequality between the raw graph's edge count and $s_d k$ is needed to define this decoder. The hybrid decoder selects this branch solely by the label-independent inequality $k < b_\star$; it does not test boundary stubs or closedness. On a genuine extracted view the advertised radius makes this exact. If the queried component has fewer than $b_\star$ vertices, the whole component is visible and $k = n$. If it has at least $b_\star$ vertices, connectedness implies that the radius- $(b_\star - 1)$ ball from a root endpoint already contains at least $b_\star$ vertices (or the whole component), so $k \geq b_\star$. Thus accidental star patterns in a genuine small-component word cannot affect parsing.

9 A total local decoder

A decoding function must be defined not only on codewords, but on every raw finite view. All five stages below operate solely on the supplied finite graph, labels, and local indices. We then prove in [Section 10](#) that an extracted radius- R_d view contains enough information for these operations to recover its root target.

A malformed raw view may repeat numerical IDs. To keep every parsed choice total, append the canonical visible local vertex index to every vertex-ID key. Thus the parsed assignment key is $(\text{dist}(v, c), \text{id}(c), \text{loc}(c))$, and the parsed edge key is its unordered endpoint-ID pair followed by the unordered local-endpoint pair. Pairs here are listed in increasing order and compared lexicographically. All distances in the parser are distances in the supplied induced graph, with distance $+\infty$ between its different connected components. These last coordinates are deterministic tie-breaks only: on a genuine extracted view the IDs are injective, so the orders reduce to the encoder's ID and κ orders. The record key in [Section 7.1](#) uses the same parsed-center convention.

The decoder has the following five stages.

1. **Select the component branch.** Let k be the total number of visible vertices. If $k < b_\star$, regard the whole visible induced graph and its labels as one finite codeword, apply the rank/unrank decoder of [Section 8.2](#), and return its queried-edge bit (or zero when the label rank is outside the target interval). Otherwise enter the parsed large-component branch.
2. **Parse stars and cells.** On the large branch, call a visible vertex a parsed marked center when its label is 1 and every one of its neighbors in the visible induced graph has label 0; opaque outside port stubs are not inspected. Define

$$\gamma(v) = \arg \min\{(\text{dist}(v, c), \text{id}(c), \text{loc}(c)) : c \text{ marked and } \text{dist}(v, c) \leq \eta\}, \quad (9.1)$$

and put $\gamma(v) = \perp$ if the set is empty. For a marked center c , let $\widehat{B}_c = \{v : \gamma(v) = c\}$. Its members lie within parsed distance η of c , but an arbitrary raw view has no degree bound, so a separate size guard is essential. Call the cell *valid* iff it contains c and every visible neighbor of c , its induced graph is connected,

$$b_\star \leq |\widehat{B}_c| \leq B_{\max},$$

and c has exactly d visible neighbors. For each valid cell set $B := \widehat{B}_c$ and construct its buddy matching and the quotient-independent quantities N_B , c_B , π_B , and σ_B .

Invalid cells expose the all-zero logical word and no nonempty record. The size and degree guards are automatic for completely reconstructed genuine encoder cells, not for arbitrary raw views. For a valid cell, $b \geq b_*$ and the center's d neighbors ensure $b - d - 1 \geq 0$; the arithmetic proof of [Lemma 4.3](#) gives $\sigma_B \geq 0$ even if other raw vertices have excessive degree.

3. **Build the quotient and decode fields.** Construct the parsed quotient on all marked centers. An original edge with a $\perp$ endpoint is omitted; an edge whose endpoints map to the same marked center is internal; every other edge becomes a quotient edge. Parallel cancellation and incidence splitting include invalid marked cells, but such cells contribute no records. The residual degree now determines ν_B , w_B , t_B , and ℓ_B for each valid cell. Rank its physical data word. If the word is illegal or its rank is outside the image in (6.9), set its entire logical word to zero; the parser retains no separate invalid-word flag. The parsed residual quotient is simple even on malformed inputs. Consequently its incidence splitting always produces paths and cycles of degrees one and two, so every orientation operation in the next stage is defined.
4. **Recover directions.** Apply the total residual-orientation rule of [Section 7.1](#). In particular, discard every record whose host is outside the radius- ρ_{host} inner edge ball before the component/distance filter and least-record selection. Keep the fixed opposite directions on canceled pairs.
5. **Find the owner and read payload.** For a queried original edge e , return zero immediately if an endpoint maps to $\perp$ or if either relevant parsed cell is invalid. Otherwise, internal ownership or the parsed quotient direction determines an eventual owner B . Enumerate by the parsed edge key above all internal edges of B and all parsed quotient edges directed out of B , omitting $\perp$ -endpoint edges exactly as in the quotient. If this list has more than π_B entries, return zero. Otherwise return the payload coordinate at the rank of e in this list. A malformed physical word has already become the all-zero logical word at Stage 3.

These conventions also cover missing records, excess records, disagreeing records, unsafe boundary-host records, extra stars, disconnected parsed cells, and owner overload. On a malformed labeling, separately decoded residual edges need not form a balanced orientation, but every edge receives a deterministic absolute direction, and the overload guard prevents an out-of-range payload access. Thus the decoder is total.

There is no circularity in the definition: star patterns determine cells; cells and the visible graph determine buddy matchings, the quotient, and field lengths; the finite code reveals records; records determine ownership; only then is the payload interpreted. We prove exact recovery next.

10 Correctness, locality, and the explicit radius

Fix a genuine encoded view $X = \text{View}_{R_d}(G, \text{id}, \ell, e)$ whose root lies in a large component. We identify its local vertices with the corresponding ambient vertices using their distinct IDs. Put

$$g = 2\eta + 2, \quad \zeta = 2\eta + 1. \quad (10.1)$$

If two global blocks are adjacent in the quotient, their centers have graph distance at most ζ : follow a path of length at most η from the first center to a witnessing boundary endpoint,

cross that edge, and follow at most η further edges to the second center. The same estimate holds for parsed cells, with distances in X , because (9.1) assigns a vertex only to a center within distance η . Thus

$$\text{one global or parsed virtual step moves its host center by at most } \zeta. \quad (10.2)$$

This bound is valid even when successive virtual copies have hosts already visited by the walk.

Recall $\rho_{\text{host}} = \eta + (2L + 1)\zeta$. The needed physical radius is

$$r_{\text{lit}} = \rho_{\text{host}} + g = \eta + (2L + 1)(2\eta + 1) + (2\eta + 2). \quad (10.3)$$

Direct expansion gives

$$R_d - r_{\text{lit}} = \eta + 2L + 3 > 0. \quad (10.4)$$

In particular, every radius- g neighborhood of a vertex within distance ρ_{host} of e is wholly present in X .

We will repeatedly use two elementary facts about induced balls. Distance to the root edge is unchanged by extraction: a shortest path to an endpoint of e stays in the edge ball. More generally, if $\text{dist}_G(e, c) + u \leq R_d$, every ambient path of length at most u starting at c lies in X . Consequently its radius- u neighborhood, including all distances at most u from c , is reproduced exactly. These facts follow directly from the triangle inequality and the definition of $V_{R_d}(G, e)$.

Lemma 10.1 (Exact reconstruction at an interior center). *A parsed marked center c with $\text{dist}_X(e, c) \leq \rho_{\text{host}}$ is a genuine encoder center. Conversely, every encoder center satisfying this distance bound is parsed as marked. At each such center the following data are exact:*

- (i) *its entire cell B_c , its internal edges, and its buddy matching;*
- (ii) *every boundary edge of B_c and the assigned center of its outside endpoint;*
- (iii) *its residual quotient incidences, their order, their pairing, the local copy indices, and the reference incidence at each copy;*
- (iv) *its logical field lengths, its payload, and every stored record.*

In particular its parsed cell is valid. If a residual edge joins two such interior centers, its two virtual endpoints agree with the global ones.

Proof. All neighbors of c are visible, so its parsed star test is the global star test. By Lemma 4.2, that test identifies precisely the encoder centers.

Every vertex that could belong to B_c is within distance η of c , by Lemma 4.1. To decide the assignment of any such vertex v , it suffices to inspect centers within distance η of v . All these potential centers are within 2η of c , and all their neighbors are within $2\eta + 1$. Their stars, the distances relevant to the assignment, and the ID comparisons are therefore exact. No spurious boundary star can compete in this assignment. Moreover a parsed member of $\hat{B}_c$ must itself be within η of c , so there are no additional parsed members outside this candidate region. Hence $\hat{B}_c = B_c$. The induced cell graph and its ordered greedy matching are then identical. The global cell geometry and regularity imply all the validity guards.

All endpoints of edges incident with B_c are within $\eta + 1$ of c . Their assignments require at most another $\eta + 1$ hops, hence lie in the available radius $g = 2\eta + 2$ neighborhood. This

proves (ii), including the absence of $\perp$ endpoints. Every parallel quotient edge incident with B_c is among these edges. Thus the complete parallel lists, their cancellations, the remaining neighbor-center keys, and all incidence pairings at B_c are exact. This proves (iii). Notice that this argument needs the center assigned to a neighboring boundary endpoint, not a reconstruction of that neighbor's entire cell.

The cell size, center degree, matching, residual degree, physical labels, and their orders now coincide. Therefore the code rank, the field lengths in (4.10) and (6.1), and all decoded fields coincide as well. The encoder's rank lies in the reserved interval, so the zero fallback does not apply. This proves (iv). At two interior endpoint centers, (iii) identifies both copy indices of a residual edge, proving the last assertion. $\square$

Lemma 10.2 (Comparison of bounded virtual walks). *Let B be an encoder block whose center is within distance η of e , and let z be any virtual copy above B . Every global or parsed virtual walk of length at most $2L + 1$ starting at z has the same sequence of named copies and residual edges in the other graph. The same assertion holds for a parsed walk of length $2L + 2$ if its terminal host is host-safe.*

Proof. By (10.2), the host center of the vertex at step i is within distance $\eta + i\zeta$ of e , in the appropriate physical graph. For $i \leq 2L + 1$ this is at most ρ_{host} . Thus all these hosts satisfy Lemma 10.1. Starting with the exact local copy at B , its residual edge and the local index of the next copy are identical whenever both endpoint hosts are interior. Induction on the walk length proves the assertion in either direction. For the extra terminal step all earlier hosts are interior by the same bound, and the terminal host is interior by its explicit safety hypothesis. Apply the last assertion of Lemma 10.1 to that step. $\square$

Lemma 10.3 (The short-component test is correct near the query). *Under the hypotheses of Lemma 10.2, if the global component of z has fewer than L vertices, its entire parsed component is the same component. If the global component has at least L vertices, the parsed component has at least L vertices.*

Proof. In the first case every vertex of the global component is reached from z by a path of length at most $L - 2$. All these vertices and edges are reconstructed by Lemma 10.2. A parsed edge leaving this set would extend such a path by one step, still well within $2L + 1$; the same lemma would give a global edge leaving the global component, a contradiction. Therefore there are no extra parsed vertices.

For the second case, a connected graph with at least L vertices has at least L vertices within distance $L - 1$ of any fixed vertex: either this ball is the whole component, or a shortest path to a vertex outside it already contains L vertices inside it. Choose L distinct such vertices and paths to them. The walk lemma embeds these paths into the parsed graph. The named-copy correspondence is injective, since a center ID and its local copy index uniquely identify a vertex. Hence the parsed component contains at least L vertices. $\square$

Lemma 10.4 (Correctness of recorded orientations). *Let B be an encoder block whose center is within distance η of the original root edge e . For every residual edge incident with B , the rule of Section 7.1, run in X , returns the encoder's direction of that edge.*

Proof. Let f be such a residual edge and let z be its virtual endpoint above B . The other host center is at most one quotient step away, so both virtual endpoints of f are exact. If the global component is short, Lemma 10.3 identifies its whole parsed component. The canonical edge order and the endpoint keys agree, so its canonical orientation agrees.

Suppose the global component is long. The same lemma says the parsed component has at least L vertices, so the decoder takes its record branch. We prove both that every retained candidate is correct and that a candidate exists.

For a retained candidate, take a parsed walk of length at most $2L + 1$ from the anchor $a(f)$ to its named copy. Prepend f if needed to start at z . The resulting walk has length at most $2L + 2$, and its terminal host is safe by the record filter. By [Lemma 10.2](#), it is a global walk with the same named copies. By [Lemma 10.1](#), the terminal record is the exact readback of an encoder slot, including its reference incidence and direction bit. Every nonempty encoder slot describes the chosen orientation of its named copy's global component. The global walk shows that this is the component containing f . Propagating that direction back along the walk therefore gives the encoder's direction of f .

Conversely, [Lemma 7.3](#) supplies a recorded copy at distance less than $2L$ from one endpoint of the global edge f . Since distances are integral, it can be reached from z in at most $2L$ steps, including a possible initial traversal of f . Every host on that walk is interior; the walk and the record are reconstructed exactly by [Lemmas 10.1](#) and [10.2](#). Its host is safe. In the parsed graph the anchor $a(f)$ is at most one step from z , so the record is within $2L + 1$ of that anchor and is eligible. Thus the candidate set is nonempty, all its elements describe the same global direction, and its least element gives the correct direction. $\square$

Lemma 10.5 (Exact recovery). *On every labeling produced by the encoder, the radius- R_d decoder returns x_e at every original edge e .*

Proof. The visible-count branch is correct by [Section 8.2](#), since $R_d \geq b_\star - 1$. On a small component, base- α encoding and binary digit extraction are inverse.

Now let e lie in a large component. Each block containing an endpoint of e has its center within distance η of e . Its cell, validity guards, and incident quotient data are exact by [Lemma 10.1](#). If e is internal, its owner is therefore the correct common block. If e is in a canceled parallel pair, the identical edge list and center order give its correct direction. Otherwise [Lemma 10.4](#) gives its correct residual direction. In all cases the decoder chooses the encoder's owner, say B .

We must also recover the position of e in B 's payload, not just its owner. The reconstruction lemma supplies every internal and boundary edge of B with the correct outside assignment. Canceled directions agree, and [Lemma 10.4](#) applies separately to every residual incidence of B . Thus the entire parsed list of edges owned by B equals its global list, including its order by κ . No incident edge is omitted for a $\perp$ endpoint. By [Lemma 5.1](#) the list has at most π_B elements, so the overload guard is inactive. Finally the logical word is read back exactly, and its payload coordinate for e is precisely the stored bit x_e . $\square$

Every operation of the decoder is a finite computation on the supplied raw view, including on malformed input. The proof uses exact ambient reconstruction only at the interior centers specified above; it makes no claim that an arbitrary cell at the outer boundary is genuine. Because the input is just the radius- R_d view, the reconstruction and recovery lemmas establish the advertised locality. For fixed d , genuine views have bounded size, while arbitrary raw structures may have larger size. The parser still terminates on every finite raw structure. No bound on its computational running time is part of the theorem.

The radius is polynomial in d . For $d \geq 3$,

$$B_{\max} \leq 3(d-1)^\eta, \quad \chi_d = O(\eta \log d) = O(d^2 \log d).$$

Indeed $d/(d-2) \leq 3$ gives the first inequality from (2.4); the second follows by taking logarithms and using $\eta = 128(d+1)^2 - 2$. Hence $L = O(d^3 \log d)$ and

$$R_d = O(d^5 \log d). \quad (10.5)$$

For cubic graphs,

$$\begin{aligned} b_\star &= 1024, & \eta &= 2046, & B_{\max} &= 3 \cdot 2^{2046} - 2, \\ \chi_3 &= 2050, & L &= 123000, & R_3 &= 246003 \cdot 4094 = 1,007,136,282. \end{aligned} \quad (10.6)$$

Here $2^{2049} < 3B_{\max} < 2^{2050}$ verifies the logarithm ceiling without numerical approximation. These constants are sufficient, not claimed to minimize the radius.

11 Order normalization and reversible strip maps

We turn to the lower bound in even degree. At the putative rate $d/2$, vertex labels and edge targets have exactly the same cardinality on every d -regular graph. Hence surjectivity would make every finite global decoder map bijective. The argument below converts these finite bijections into a reversible map on a one-dimensional strip.

The decoder may initially use the numerical values of IDs, whereas reflection only respects their order. The following exact normalization removes this issue. It is a direct Ramsey argument specialized to our surjectivity model.

Lemma 11.1 (Order normalization). *Fix d, r and a finite alphabet. Suppose a radius- r decoder is surjective on every finite simple d -regular graph for every injective natural-number ID assignment. There is a decoder of the same radius and alphabet which is still surjective on that family and depends on IDs only through their relative order in the queried view.*

Proof. One may take $q = 2 \sum_{i=0}^r d^i$ as a bound on the number of visible vertices: at most d^i walks of length i start at either endpoint. For each $1 \leq j \leq q$, combine into one finite color the original decoder's complete truth table on every j -vertex rooted topology of maximum degree at most d , every port template with d entries at each vertex, and every physical-label assignment when a given j -set of IDs is placed increasingly on the ordered template positions. There are finitely many templates and labels, so the color set is finite.

We recall the short proof of the finite-color infinite Ramsey theorem. It is by induction on the subset size. The singleton case is infinite pigeonhole. For the step from j to $j+1$, recursively choose points z_i and nested infinite tails so that, after fixing z_i , the color is constant on all j -subsets of the next tail; this uses the induction hypothesis. Infinitely many z_i have the same resulting color. Any $j+1$ of those points has the color attached to its least point, so they form an infinite homogeneous set.

Apply this theorem successively for $j = 1, \dots, q$, restricting the current infinite set after each application. We obtain one infinite ID set I homogeneous for every relevant view size. On a determining view of size at most q with a strict ID order, define the new decoder to be the common truth-table value obtained from increasing IDs in I ; on every other raw view give it an arbitrary fixed output, so it remains total. For a finite instance, relabel the vertices order-preservingly into I . The given decoder has a valid encoding there. Copy its physical labels back. Homogeneity preserves every local truth-table value, and the order-preserving relabeling also preserves neighbor-ID ports. Thus the new decoder realizes the original target. The normalized rule ignores numerical magnitudes; the canonical view enumeration and ports still retain the relative order. $\square$

Fix an even degree $d = 2k$, $k \geq 1$, and assume for contradiction that a k -bit decoder is surjective on all finite simple $2k$ -regular graphs. By Lemma 11.1, assume it is order-invariant. Choose the positive jump lengths

$$a_1 = 1, \quad a_j = 2(j-1) \quad (2 \leq j \leq k), \quad S_k = \sum_{j=1}^k a_j = 1 + k(k-1). \quad (11.1)$$

Thus S_k is odd. Put $M = \max_j a_j$, taking $M = 1$ for $k = 1$. For $n > 2M$, define

$$G_n = \text{Cay}(\mathbb{Z}/n\mathbb{Z}, \{\pm a_1, \dots, \pm a_k\}), \quad e_{j,i} = \{i, i + a_j\}. \quad (11.2)$$

The graph is connected because $a_1 = 1$. It is simple and $2k$ -regular: the positive jumps are distinct and less than $n/2$, so none is zero, no two are congruent up to sign, and reversing a named edge cannot name it again. Consequently (11.2) names every edge exactly once.

For every total order $\prec$ on G_n , the decoder induces

$$D_{n,\prec} : (\{0, 1\}^k)^{\mathbb{Z}/n\mathbb{Z}} \longrightarrow \{0, 1\}^{E(G_n)}. \quad (11.3)$$

Both sides have 2^{kn} elements. Surjectivity therefore makes $D_{n,\prec}$ a bijection. A finite total order is realized by assigning its positions the distinct natural IDs $0, \dots, n-1$.

A *configuration* over a finite alphabet C is a function $x : \mathbb{Z} \rightarrow C$; the space $C^{\mathbb{Z}}$ of all such configurations is called the *full shift*. A map $F : C^{\mathbb{Z}} \rightarrow C^{\mathbb{Z}}$ has dependence radius m if $F(x)_i = F(x')_i$ whenever x, x' agree on $[i-m, i+m]$. Its coordinate rule may depend on i . We call it *N-periodic* when it commutes with translation by N sites, equivalently when the coordinate rules repeat every N sites.

The order-normalized decoder can be applied to the infinite jump graph with any total order: each finite view is evaluated by its relative order, or equivalently by replacing its visible IDs with their ranks. No order-preserving assignment of natural IDs to the whole infinite graph is required. Write

$$m = (r+2)M.$$

The edge $e_{j,i}$ has its endpoints in $[i, i+M]$. Its radius- r vertex view, together with the additional neighbor layer needed to determine all its ports, is contained in $[i-m, i+m]$. Thus grouping the k output bits based at i gives a map of dependence radius at most m for every fixed total order.

Lemma 11.2 (Realizing a finite fragment on a ring). *Fix any total order of the infinite jump graph and a finite interval $J = [u, v] \subseteq \mathbb{Z}$. For a sufficiently large $n > 2M$, one can order G_n so that all decoder outputs based in J agree with the infinite ordered graph, for every assignment of labels on $[u-m, v+m]$ copied to the corresponding ring vertices.*

Proof. Choose $n > 2(v-u+2m+2M+1)$. Projection modulo n is injective on $I = [u-m, v+m]$. For two coordinates in I , a difference congruent to a jump $\pm a_j$ modulo n must equal that jump as an integer: the size of n excludes an additional nonzero multiple of n . Hence projection preserves the induced edges on I . A walk of at most $r+1$ jumps starting at an endpoint of $e_{j,i}$, $i \in J$, stays in I . All determining views and their neighbor layers are therefore reproduced, with no additional vertices or shortcuts from wrapping around the ring. Impose the given order on the projected vertices of I and extend it arbitrarily to the other ring vertices. Local order comparisons and neighbor-ID port positions are identical. Since labels outside I are not read by these queries, all the specified outputs agree. $\square$

Lemma 11.3 (Finite-alphabet compactness and inverse coordinates). *Let C be a finite nonempty alphabet and let $F : C^{\mathbb{Z}} \rightarrow C^{\mathbb{Z}}$ have finite dependence radius. Every sequence of configurations has a subsequence that eventually stabilizes at each coordinate. The image of F is closed under these coordinatewise limits. In particular, if every finite interval of every desired output can be realized, F is onto. If F is bijective, then for each i there is a finite t_i such that $F^{-1}(y)_i$ is determined by $y|_{[i-t_i, i+t_i]}$.*

Proof. List the coordinates as $0, 1, -1, 2, -2, \dots$. Repeatedly pass to an infinite subsequence constant at the next coordinate, using finiteness of C . The diagonal subsequence stabilizes at every coordinate. A finite-radius map preserves this convergence because each output coordinate reads only finitely many input coordinates.

If $F(x^{(n)})$ converges coordinatewise to y , extract a convergent subsequence of the inputs with limit x . Then $F(x) = y$, proving closedness. If every interval $[-n, n]$ of y has a preimage, choose one such input for each n and apply the same argument; the image limits to y , proving surjectivity.

Finally suppose no t_i works for a fixed i . For each n choose outputs $y^{(n)}, z^{(n)}$ agreeing on $[i - n, i + n]$ whose preimages differ at i . Apply the diagonal argument to the pairs of preimages, over the finite alphabet $C \times C$. Their limits x, x' still differ at i , since the pair of symbols at i eventually stabilizes. For each output coordinate j , finite dependence and the growing agreement intervals give $F(x)_j = F(x')_j$. This contradicts injectivity. $\square$

Lemma 11.4 (Periodic finite bijections give a reversible map). *Let C be a finite nonempty alphabet, let $N \geq 1$, and let $F : C^{\mathbb{Z}} \rightarrow C^{\mathbb{Z}}$ have a finite dependence radius and an N -periodic local rule. If F restricts to a bijection on the configurations whose period divides tN for every $t \geq 1$, then F is bijective and F^{-1} has a uniform finite dependence radius.*

Proof. Any prescribed output word on a finite interval extends to a tN -periodic configuration for a sufficiently large t . The assumed bijection supplies a same-period preimage. Thus [Lemma 11.3](#) makes F onto.

For injectivity, group consecutive N sites into single symbols of $A = C^N$. The blocked map has a translation-invariant finite local rule; choose a radius $r_0 \geq 1$ for that rule. A pair of blocked input configurations has equal outputs precisely when each pair window of length $2r_0 + 1$ gives equal central outputs. This condition has the following explicit finite graph representation. Its vertices are all allowed length- $(2r_0 + 1)$ words over $A \times A$, and a directed edge joins two such words when their last and first $2r_0$ entries agree. Bi-infinite walks give exactly the equal-output pairs, by reading the central entries of successive windows.

Suppose two distinct infinite inputs have equal outputs. If they differ at infinitely many blocked coordinates, there are infinitely many window vertices with unequal central entries. Some such vertex occurs at two different positions, because the graph is finite. Repeating the directed segment between these positions produces a periodic walk with an unequal central entry. Reading its centers gives two distinct periodic inputs with equal outputs. Their unblocked period is a multiple of N , contradicting one of the finite-period bijections.

Otherwise the difference is contained in a finite blocked interval $[a, b]$. Choose $t > b - a + 4r_0 + 1$. On a ring of t blocks, copy both inputs on $[a - 2r_0, b + 2r_0]$ and give them a common fixed symbol elsewhere. Their difference is still nonempty and is confined to $[a, b]$. Every output window meeting this difference is contained in the copied interval and has the equal outputs of the original collision. All other output windows see identical inputs. Equivalently, periodically repeating the two ring words gives a collision of period tN , again impossible.

Thus F is bijective. By [Lemma 11.3](#), each inverse coordinate has a finite dependence interval. The inverse commutes with translation by N , because F does. Take the maximum of the dependence radii at the N representatives $0, \dots, N-1$; translation then gives the same finite bound at every coordinate. $\square$

Choose $N > 4(r+2)M$. Projection modulo N is then injective on every determining view in the infinite jump graph, including the neighbor layer which fixes boundary ports. Fix any ranking π of the residues modulo N . On $\mathbb{Z}$, order $i = u + tN$, $0 \leq u < N$, lexicographically by $(\pi(u), t)$. No determining view contains two positions with the same residue, so its comparison pattern is N -periodic.

On G_{tN} , use the finite keys $(\pi(u), v)$, $0 \leq v < t$, and replace them by their ranks. Every local comparison is decided by the distinct primary residue ranks, including at the global wrap. Hence topology, order, and ID-induced ports are exactly the finite projection of the infinite environment.

Group the k edge outputs based at i into one k -bit output cell:

$$P(x)_i = (D(x)_{e_{1,i}}, \dots, D(x)_{e_{k,i}}). \quad (11.4)$$

This is an N -periodic finite-radius map on the 2^k -symbol full shift. On configurations whose period divides tN , it is precisely the bijection (11.3). Therefore [Lemma 11.4](#) makes P bijective with a uniform finite-radius inverse. Grouping every N consecutive sites into one cell gives a reversible one-dimensional cellular automaton.

12 A self-contained rational flow index

We prove the one-dimensional index facts needed below by counting finite input/output interfaces. For historical context on reversible cellular automata and their information-flow invariants, see Hedlund [[Hed69](#)], Kari [[Kar96](#)], and Gross, Nesme, Vogts, and Werner [[GNVW12](#)]. The argument here requires none of their results.

Call a map *reversible and local* if it is a bijection and both it and its inverse have a uniform finite dependence radius. Two such maps agree on a negative tail if their coordinate functions agree, for every input, at all sufficiently negative sites; positive-tail agreement is defined analogously. When we say that a map has given forward and inverse rules on a tail, both assertions use this pointwise, all-input meaning.

The invariant counts what a reversible map can transmit across one cut. We record a $2r$ -symbol input word just to the right of the cut together with a $2r$ -symbol output word straddling it. Increasing r exposes exactly three new freely choosable symbols, which explains the normalization by q^{3r} below. The proof consists of making these statements into explicit finite bijections.

Lemma 12.1 (Rational flow index). *Let C be a finite nonempty alphabet and let $F : C^{\mathbb{Z}} \rightarrow C^{\mathbb{Z}}$ be a possibly nonuniform bijection. Suppose both F and F^{-1} have one uniform finite dependence radius. There is a positive rational number $\text{ind}(F)$ with the following properties.*

- (i) *If H and G are reversible local maps on the same alphabet, H, H^{-1} agree respectively with F, F^{-1} on a negative tail, and H, H^{-1} agree respectively with G, G^{-1} on a positive tail, then $\text{ind}(F) = \text{ind}(G)$.*
- (ii) *Indices multiply under composition and under independent parallel tracks.*

- (iii) *Spatial reflection reciprocates the index.*
- (iv) *For the convention $(S_m x)_i = x_{i-m}$ on an alphabet of size q , $\text{ind}(S_m) = q^m$. In particular, a binary track has index 2^m . On-site bijections, which permute symbols separately at each site, have index one.*

Proof. Write $q = |C|$ and let ρ be a common dependence radius for F and F^{-1} . For a cut $c \in \mathbb{Z}$ and $r \geq \rho$, let

$$\mathcal{I}_F(c, r) = \left\{ \left(x|_{[c, c+2r-1]}, F(x)|_{[c-r, c+r-1]} \right) : x \in C^{\mathbb{Z}} \right\}, \quad N_F(c, r) = |\mathcal{I}_F(c, r)|. \quad (12.1)$$

Intervals of length zero are empty. These are finite nonempty sets: any global configuration supplies a trace.

We first remove dependence on the chosen radius. Forgetting the outer shell of a radius- $(r+1)$ trace while retaining the two new right input symbols and the new left output symbol gives

$$\mathcal{I}_F(c, r+1) \longrightarrow \mathcal{I}_F(c, r) \times C^3, \quad z \longmapsto (z_{\text{core}}, x_{c+2r}, x_{c+2r+1}, F(x)_{c-r-1}). \quad (12.2)$$

This map is a bijection. For injectivity, the retained data give the entire larger input window and every larger output symbol except $F(x)_{c+r}$; that last symbol is determined by forward locality from the input interval $[c, c+2r]$.

For surjectivity, choose a global witness x for the prescribed core trace. Replace x_{c+2r} and x_{c+2r+1} by the two prescribed right symbols. The core outputs, whose dependence neighborhoods end at $c+2r-1$, do not change. Apply F , replace the output coordinate $c-r-1$ by the prescribed left symbol, and pull the result back through F^{-1} . Inverse locality says that this last modification changes no input coordinate at or to the right of c . Hence the pulled-back configuration realizes exactly the desired larger trace. Therefore

$$N_F(c, r+1) = N_F(c, r)q^3.$$

We next move the cut. Let $\mathcal{B}_F(c, r)$ record the two length- $(2r+1)$ words

$$x|_{[c, c+2r]} \quad \text{and} \quad F(x)|_{[c-r, c+r]}.$$

There are exact bijections

$$\mathcal{B}_F(c, r) \simeq \mathcal{I}_F(c, r) \times C, \quad \mathcal{B}_F(c, r) \simeq \mathcal{I}_F(c+1, r) \times C. \quad (12.3)$$

For the first, keep the last input symbol; forward locality determines the only omitted output symbol. Conversely, set that last input symbol in a witness for the interface trace; all recorded outputs to its left remain fixed. For the second, keep the first output symbol; inverse locality determines the only omitted input symbol. Conversely, set that first output symbol and pull back by F^{-1} ; all later recorded inputs remain fixed. Thus $N_F(c, r)q = N_F(c+1, r)q$. Since $q > 0$, cancellation and induction in both directions on $\mathbb{Z}$ show that $N_F(c, r)$ is independent of c .

Define

$$\text{ind}(F) = \frac{N_F(c, r)}{q^{3r}} \quad (r \geq \rho). \quad (12.4)$$

Equations (12.2)–(12.3) prove that this definition is independent of both r and c . It is a strictly positive rational number. This direct cut-by-cut argument has no blocking, pairing-offset, or common-coarsening assumption.

The interface formula is local in the exact sense required for (i). If the forward coordinate rules of two reversible maps agree throughout the output interval $[c - r, c + r - 1]$, their interface sets at (c, r) are equal: the same global witness realizes the same input and output words. Given maps with possibly different radii, take r to be their maximum. Agreement on a left tail ending at b permits the cut $c = b - r$, while agreement on a right tail beginning at b permits $c = b + r$. Cut and radius independence then give equal indices. Applying this on the two tails of H proves (i). Agreement of the inverse rules is part of saying that H is a reversible crossover; the trace comparison itself needs only the forward equality.

For independent maps F on C and G on D , coordinate projection is an exact bijection

$$\mathcal{I}_{F \times G}(c, r) \simeq \mathcal{I}_F(c, r) \times \mathcal{I}_G(c, r).$$

Every pair of component traces has a witness obtained by pairing their two global witnesses. Since $|C \times D|^{3r} = |C|^{3r}|D|^{3r}$, parallel indices multiply. A coordinatewise alphabet bijection similarly bijects interface traces and leaves the index unchanged. A radius-zero on-site bijection has only the pair of empty words in its radius-zero trace, so its index is one.

For composition, let F and G act on the same alphabet C . Place two copies of $C^{\mathbb{Z}}$ in parallel and let S_+ swap their two coordinates at every nonnegative site while acting identically on the negative half-line. With composition read from right to left, the reversible map

$$K = (F \times 1) \circ S_+ \circ (1 \times G) \circ S_+ \quad (12.5)$$

has a local inverse $S_+ \circ (1 \times G^{-1}) \circ S_+ \circ (F^{-1} \times 1)$. Let ρ_F, ρ_G bound the forward and inverse radii of F, G . At a site $i < -(\rho_F + \rho_G)$ every intermediate site read in either composition is negative; the swaps do nothing and the forward and inverse rules are those of $F \times G$. At $i > \rho_F + \rho_G$ every such site is positive; both swaps act, and the rules are those of $(F \circ G) \times 1$ and its inverse. Thus K is a reversible local map with the asserted two tails. Part (i), parallel multiplicativity, and the on-site value therefore give $\text{ind}(F \circ G) = \text{ind}(F) \text{ind}(G)$. Applying this identity to F and F^{-1} gives $\text{ind}(F^{-1}) = \text{ind}(F)^{-1}$.

Let $(Jx)_i = x_{-i}$ and $F^\# = JFJ$. Reversing both recorded words and exchanging their input/output roles is an exact bijection

$$\mathcal{I}_{F^\#}(c, r) \simeq \mathcal{I}_{F^{-1}}(-c - r + 1, r).$$

Indeed, set $c' = -c - r + 1$, $u = Jx$, and $y = F(u)$. Reversing the $F^\#$ -output word gives $y|_{[c', c' + 2r - 1]}$, while reversing the input word gives $u|_{[c' - r, c' + r - 1]} = F^{-1}(y)|_{[c' - r, c' + r - 1]}$. Reversing and exchanging the two words therefore sends every $F^\#$ trace to an F^{-1} trace. Conversely, given an F^{-1} trace witnessed by y , set $u = F^{-1}(y)$ and $x = Ju$; the same coordinate calculation in reverse sends it to an $F^\#$ trace. These two constructions are inverse, so the displayed correspondence is an exact bijection. Cut independence and the inverse identity show $\text{ind}(F^\#) = \text{ind}(F)^{-1}$, proving (iii).

Finally consider $S_1 x_i = x_{i-1}$. At $c = 0, r = 1$, its trace consists of the four freely chosen symbols x_0, x_1, x_{-2}, x_{-1} . Thus $N_{S_1}(0, 1) = q^4$ and (12.4) gives $\text{ind}(S_1) = q$. Composition and inversion now give $\text{ind}(S_m) = q^m$ for every $m \in \mathbb{Z}$; taking $q = 2$ proves (iv). $\square$

13 The even-degree lower bound

Reflect the infinite jump graph by $\rho(i) = -i$, transport the total order, and let Q be the resulting decoder map. On vertex cells put $(Rx)_i = x_{-i}$. Graph reflection sends

$$\rho(e_{j,i}) = \{-i, -i - a_j\} = e_{j, -i - a_j}. \quad (13.1)$$

Thus order invariance gives the exact coordinate identity

$$Q = R_E P R, \quad (R_E y)_{j,i} = y_{j,-i-a_j}. \quad (13.2)$$

Let R_0 reflect every output track without displacement and define

$$(U y)_{j,i} = y_{j,i+a_j}. \quad (13.3)$$

Then $R_E = U R_0$. The map $R_0 P R$ and its inverse have the same dependence bounds as P and P^{-1} ; U and U^{-1} have radius at most M . Consequently $Q = U(R_0 P R)$ is reversible and local. If $\alpha = \text{ind}(P)$, [Lemma 12.1](#) and the fixed convention $(S_m x)_i = x_{i-m}$ give

$$\text{ind}(Q) = \frac{2^{-S_k}}{\alpha}. \quad (13.4)$$

Indeed, $R_0 P R$ is the spatial reflection of P , while U shifts binary track j by $-a_j$ in this convention, and independent tracks multiply.

It remains to force the two periodic maps to have the same index. Choose one total order on $\mathbb{Z}$ which has the original periodic comparison pattern sufficiently far left and the reflected pattern sufficiently far right. For example, retain the two tail orders, choose any order on a finite transition set, and take the ordered sum of the three sets. Every decoder view far inside a tail lies wholly in that tail. Running the decoder in this ordered jump graph gives a uniformly forward-local map

$$H : (\{0, 1\}^k)^{\mathbb{Z}} \longrightarrow (\{0, 1\}^k)^{\mathbb{Z}}$$

which equals P on a negative output tail and Q on a positive output tail.

We check that H is a reversible crossover, rather than merely a surjective map.

Surjectivity. Fix a desired output y and a finite interval J . By [Lemma 11.2](#), the spliced order on all determining views for J can be realized in a sufficiently large ordered G_n . Assign to its $k|J|$ corresponding edges the desired bits and complete the finite target arbitrarily. Finite-graph surjectivity supplies labels. Copy the labels on the determining fragment back to the line and fill the other sites arbitrarily. The resulting input has H -output $y|_J$. Since this works for every J , [Lemma 11.3](#) proves that H is onto.

Injectivity. Suppose $H(x) = H(x')$. Let b_- be a left-tail bound with $H(z)_j = P(z)_j$ for all inputs z and all $j \leq b_-$, and let ρ_P be an inverse radius of P . For $i \leq b_- - \rho_P$, the output interval used by P^{-1} lies in that tail. Applying its local rule to the equal H -outputs gives $x_i = x'_i$. The corresponding argument with the right-tail bound and the inverse of Q gives equality sufficiently far right.

Any nonempty difference is therefore contained in a finite interval $[a, b]$. The forward radius bound m shows that only outputs based in $J = [a - m, b + m]$ can be affected. Apply [Lemma 11.2](#) to this interval and its determining fragment $[a - 2m, b + 2m]$, taking the ring large enough that these intervals and their radius- m neighborhoods do not identify distinct coordinates. Copy both labelings on the determining fragment and make them identical elsewhere on the ring. Outputs based in the projection of J agree because they reproduce the line collision. Every other output has identical inputs in its determining neighborhood: a coordinate within m cyclic steps of the difference would already be in the projection of J . Thus these two distinct ring labelings have equal outputs, contrary to the bijection (11.3).

Hence H is bijective. To check uniform inverse locality, first observe that its inverse agrees with P^{-1} on a negative tail. Indeed, for arbitrary $y = H(x)$ and $i \leq b_- - \rho_P$, the entire

interval read by P^{-1} sees the same values in y and $P(x)$, and so $P^{-1}(y)_i = x_i = H^{-1}(y)_i$. The same argument gives agreement with Q^{-1} on a positive tail. By [Lemma 11.3](#), each of the finitely many remaining middle inverse coordinates has a finite dependence interval. Taking the maximum of their radii and the two tail inverse radii gives one finite radius for H^{-1} . Thus H is a reversible local crossover.

Property (i) of [Lemma 12.1](#) now gives

$$\text{ind}(P) = \text{ind}(Q). \quad (13.5)$$

Combining (13.4) and (13.5),

$$\alpha^2 = 2^{-S_k}. \quad (13.6)$$

This is impossible for a positive rational α and odd S_k . To see it directly, write $\alpha = p/q$ with positive integers p, q . Then $2^{S_k} p^2 = q^2$. Write $p = 2^u p_0$ and $q = 2^v q_0$ with p_0, q_0 odd, by dividing each integer by 2 until it becomes odd. Comparing the powers of 2 on the two sides gives $S_k + 2u = 2v$. The left side is odd and the right side is even, a contradiction.

For completeness, the ordinary cardinality bound on the concrete graph K_{2k+1} already gives $s \geq k$ for any achievable s -bit scheme. The contradiction above excludes the sole remaining equality case $s = k$.

Theorem 13.1 (Even-degree locality penalty). *For every $k \geq 1$, no k -bit fixed-radius decoder is surjective on every finite simple $2k$ -regular graph. Hence*

$$b_{\text{reg}}(2k) \geq k + 1.$$

The preceding contradiction proves the theorem. At $k = 1$, the jump set is $\{1\}$ and (13.6) specializes to $\alpha^2 = 2^{-1}$. At $k = 2$, the jumps are 1, 2 and reflection displaces three binary track-sites in total, giving $\alpha^2 = 2^{-3}$.

14 Correctness and optimality

We can now finish the main theorem.

Proof of Theorem 2.2. Fix $d \geq 3$, any finite simple d -regular graph with injective identifiers, and any target $x \in \{0, 1\}^E$. Encode each component by [Section 8](#). The decoder of [Section 9](#) is a single deterministic total rule, and [Section 10](#) shows that it uses radius at most R_d . Exact recovery is [Lemma 10.5](#). Thus $b_{\text{reg}}(d) \leq \lceil (d+1)/2 \rceil$.

If d is odd, fix the identifier assignment on K_{d+1} . A decoder with s label bits induces a map from its $2^{s(d+1)}$ vertex-label assignments to the $2^{d(d+1)/2}$ possible edge vectors. Surjectivity requires $s(d+1) \geq d(d+1)/2$. Since s is integral,

$$s \geq \left\lceil \frac{d}{2} \right\rceil = \frac{d+1}{2}.$$

If $d = 2k$ is positive and even, the same cardinality argument first gives $s \geq k$, and [Theorem 13.1](#) rules out equality. Hence $s \geq k + 1 = \lceil (d+1)/2 \rceil$. This proves the claimed equality for every $d \geq 3$.

When $d = 1$, every component is an isolated edge. Give its lower-identifier endpoint the target bit and its higher-identifier endpoint any fixed bit. The radius-zero edge decoder reads

the lower endpoint. The same counting argument on K_2 gives the matching one-bit lower bound. When $d = 2$, orient each edge from its lower-identifier endpoint to its higher-identifier endpoint. Give each vertex two bits, one in each port slot, store an edge’s target bit in its slot at the tail, and fill the other slots with zero. The radius-zero decoder reads that slot, with the deterministic raw-view convention specified in [Section 2](#). The matching two-bit lower bound is [Theorem 13.1](#) with $k = 1$. This proves (2.7), and (2.8) is the arithmetic in (10.6). Finally, degree zero has no edges, so the zero-bit alphabet and the constant-zero decoder suffice; a smaller label width is not a nonnegative integer. Hence $b_{\text{reg}}(0) = 0$. $\square$

All operations in the construction are componentwise, so disconnected graphs cause no coupling. The proof assumes neither expansion nor bridgelessness. A large one-block component has $\delta_B = 0$ and no records; the handshake lemma makes $d|B|$ even, so $|E(B)| = d|B|/2 = \pi_B$ and its edge bits fit exactly in the payload. The result is therefore uniform across all component and block shapes.

Finally, the proof is logically independent of the cited decompression and index theorems. The citations establish context and provenance. The finite block code, quotient reduction, orientation records, Hall argument, total decoder, physical radius simulation, order normalization, periodic-ring reduction, finite-interface index, and crossover calculation were all defined and proved above.

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